

JAGAN THOZHUTHUNGAL JIJU

Applied Machine Learning & Search Intelligence Practitioner

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PROFESSIONAL SUMMARY

Applied Machine Learning practitioner specializing in tabular ranking systems, predictive data modeling, and search intelligence. Experienced with large-scale search datasets (Search Console, GA4), building end-to-end feature engineering pipelines with strict anti-leakage quarantines, and evaluating models with client-holdout cross-validation. Demonstrated track record of beating heuristic baselines by 2.8x on held-out domains. Certified in Anthropic AI Fluency.

TECHNICAL SKILLS

Core Languages & Libraries:	Python, Pandas, NumPy, Scikit-Learn, DuckDB, SQL, ReportLab, Matplotlib, Scipy
Machine Learning & Systems:	Tabular Ranking, Precision@K Optimization, Client-Holdout Cross-Validation, Leakage Auditing, Heuristics
Software Engineering & Tools:	Git, GitHub Actions (CI/CD), Pytest (Data Contracts), Jupyter Notebooks, Hugging Face Hub

EXPERIENCE & KEY PROJECTS

Machine Learning & Search Intelligence Intern — FlyRank	Jan 2026 – Present
Content Refresh Ranking Engine: Formulated, engineered, and evaluated a predictive ranking model to triage 30,000+ decaying URLs across 32 client domains where 54.2% of pages were slipping in search visibility. Precision Uplift (2.83x): Outperformed the industry-standard heuristic baseline rule under strict client-holdout cross-validation, achieving Precision@50 of 0.680 vs baseline 0.240 and ROC-AUC of 0.747. Zero Data Leakage Protocol: Designed strict feature isolation boundaries preventing circular outcome variables (e.g. <i>trend_pct</i>) and legacy business flags (<i>health_score</i>) from leaking into training matrices. Deployed Research Paper: Authored and published an open-source technical research paper on GitHub Pages detailing data contracts, failure modes, false-positive audits, and an operational triage playbook.	
Data Contracts & Anti-Leakage CI Harness — Open Source	Feb 2026
Pipeline Integrity Suites: Authored comprehensive pytest suites enforcing strict invariant checks (NaN prevention, zero-division bounding, and rate-scaling verification) across 52 engineered search features. Automated CI Smoke Tests: Configured GitHub Actions CI workflows that block data leaks, enforce headless script reproducibility, and guarantee zero model drift.	

EDUCATION & CERTIFICATIONS

Bachelor of Technology / Science in Computer Science / Engineering	Graduation: 2026
Anthropic Academy Certification: <i>AI Fluency: Framework & Foundations</i> (100% Completed, Verified 2026). Specialized in prompt engineering architectures, task delegation frameworks, and model evaluation. Key Coursework: Data Structures & Algorithms, Machine Learning, Database Management Systems, Statistics & Probability.	